# Network Intrusion Detection System

Real-Time Network Monitoring and Threat Detection

BCSE308L | REVIEW-2

COMPUTER NETWORKS PROJECT


## Team Members

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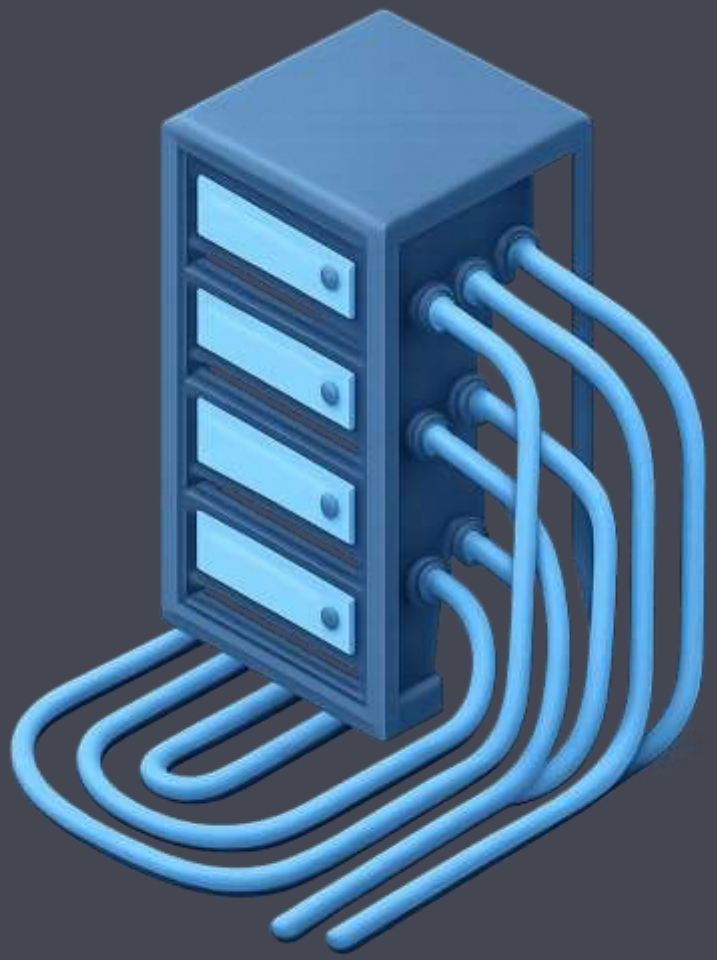
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## Submitted To

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# Introduction

The rapid expansion of computer networks and Internet-based applications has dramatically increased the attack surface for malicious actors. Traditional firewalls control traffic flow but lack the granular visibility needed to detect sophisticated threats in real time.

## Growing Networks

Rapid growth of computer networks and Internet-based applications intensifies security concerns across all environments.

## Modern Threats

Port scanning, TCP SYN flood attacks, unauthorized access attempts, and abnormal network behavior are increasingly common.

## Firewall Limitations

Traditional firewalls control traffic but lack detailed visibility into suspicious activities occurring within the network.

## IDS Solution

Continuously monitors network traffic and identifies potential security threats in real time using Python and Scapy for live packet capture, with a Flask-based web dashboard for visualization.

Key Concepts: TCP/IP protocol stack, IP addressing, port numbers, and real-time network monitoring.

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# Existing Systems

A survey of established IDS solutions reveals a consistent trade-off between detection capability and deployment complexity — particularly for small-scale and educational environments.

| System | Approach | Strength | Limitation |
|---|---|---|---|
| Snort IDS | Signature-based detection using predefined rules | Widely used; detects port scans, protocol violations, malware | Complex rule configuration for beginners and small-scale environments |
| Suricata IDS | Deep packet inspection and protocol analysis | Analyzes high-speed network traffic effectively | Requires more computational resources and configuration overhead |
| Zeek Monitor | Generates structured logs describing network activities | Useful for analyzing large networks in detail | Complex deployment and configuration for educational projects |
| Signature-Based | Compares traffic with known attack patterns | Effective for previously known attacks | Cannot detect new or unknown attack patterns |
| ML-Based IDS | Analyzes traffic patterns; classifies normal vs. malicious | Can detect unknown attacks based on abnormal behavior | Requires large datasets, training, and high computational resources |

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# Problem Statement

## The Core Challenge

Networks face unauthorized access, port scanning, DoS attacks, and abnormal activity — yet traditional security mechanisms lack continuous visibility and real-time alerting capability.

Existing IDS solutions are complex and resource-intensive, making them unsuitable for small networks and educational environments.

## Solution Requirements

A cost-effective, real-time IDS must be capable of:

01

### Capture

Capturing live network traffic from the network interface

02

### Analyze

Analyzing packets in real time for suspicious patterns

03

### Detect

Detecting suspicious activities such as port scans and SYN floods

04

### Alert

Generating immediate alerts with attack details and severity

05

### Monitor

Providing centralized monitoring through a web dashboard

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# Proposed System

A lightweight, Python-based IDS that captures, inspects, classifies, and visualizes network threats in real time — purpose-built for small networks and educational environments.



## Live Traffic Capture

Scapy captures packets from the network interface, extracting source/destination IP, protocol, ports, TCP flags, and timestamp.



## Threat Detection

Port scan detection identifies multiple connection attempts to different ports. SYN flood detection flags large SYN packet volumes within short periods.



## Severity Classification

Threats are categorized as **Low**, **Medium**, or **High** severity. Alerts are generated with full attack details and stored for future analysis.



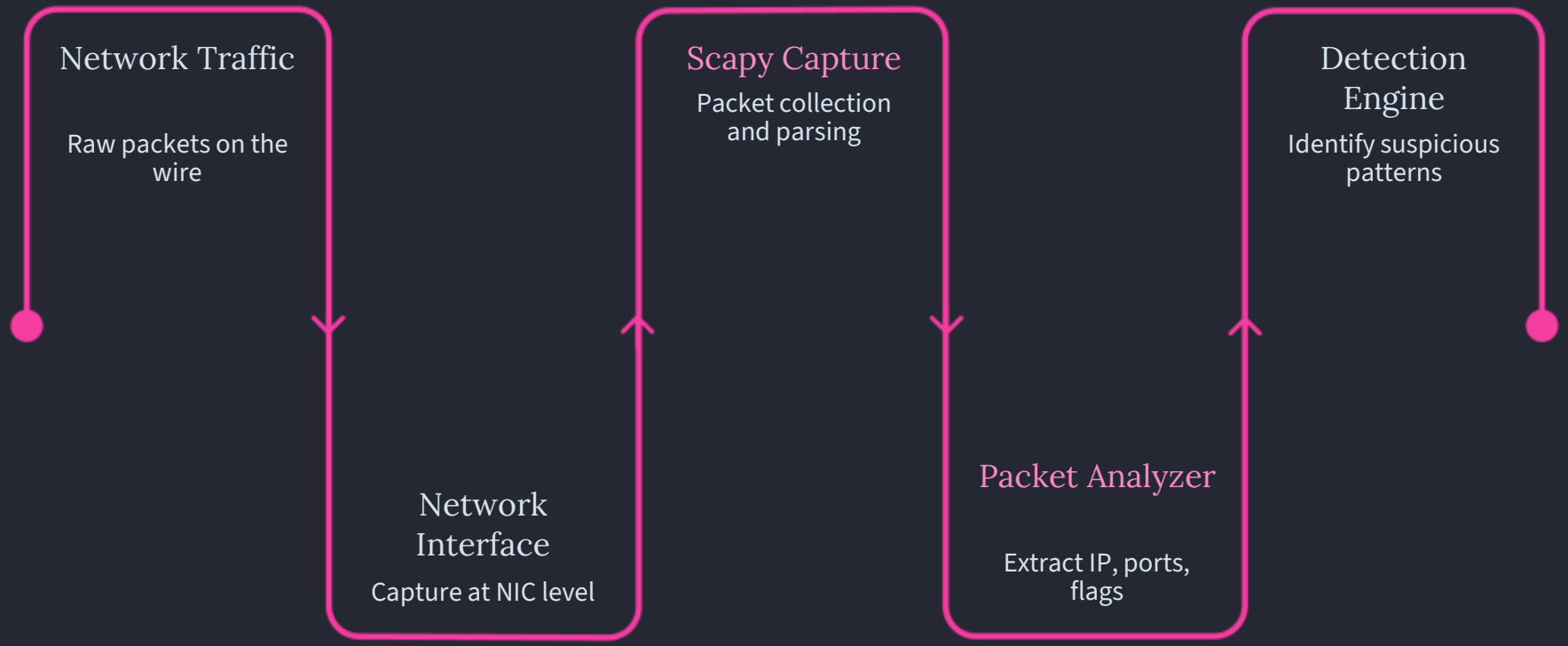
## Real-Time Dashboard

Flask server processes IDS data. Web-based interface displays network statistics, active threats, and security alerts to the administrator.

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# Proposed Architecture

A ten-stage data flow pipeline from raw network traffic to administrator-facing security dashboard.




**Capture Layer**

Network Interface + Scapy collect raw packets

**Analysis Layer**

Packet Analyzer extracts IP, ports, flags, timestamp

**Detection Layer**

Detection Engine applies rules; Alert Generator fires on threats

**Presentation Layer**

Flask Server + Web Dashboard deliver insights to Administrator

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# Novelty of the System

Six distinguishing innovations set this IDS apart from conventional signature-only or ML-heavy approaches.

1

## Hybrid Detection Approach

Combines packet-level inspection with behavior-based detection, analyzing traffic patterns over time rather than relying solely on static signatures.

2

## Real-Time Threat Severity Classification

Categorizes every detected threat as **Low**, **Medium**, or **High** based on the level and frequency of suspicious activity.

3

## Dynamic Threat Score

Calculates a cumulative threat score for each IP address based on the number and severity of suspicious events observed over time.

4

## Live Network Visualization

Dashboard displays total packets, active IPs, detected attacks, threat severity distribution, and protocol statistics — all updated in real time.

5

## Automatic Suspicious IP Tracking

Maintains a persistent list of IPs with repeated suspicious activities and continuously increases their threat score with each new event.

6

## Lightweight Educational IDS

Runs on standard consumer hardware with no specialized infrastructure, making it fully suitable for educational environments and small networks.

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# Tools and Technologies

## Backend & Detection

### Python

Primary programming language for all IDS implementation logic

### Scapy

Python library for live packet capture and deep packet analysis

### Flask

Python web framework powering the server and dashboard API

### SQLite / CSV

Stores detected intrusion events and historical security logs

## Frontend & Validation

### HTML

Structures the web dashboard interface for the administrator

### CSS

Designs the visual appearance and layout of the dashboard

### JavaScript

Dynamically updates network statistics and security alerts in real time

### Wireshark

Tests and validates captured network traffic during development

### Web Browser

Provides administrator access to the real-time monitoring dashboard

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# Implementation Methodology

A nine-step sequential pipeline transforms raw network packets into actionable security intelligence displayed on the administrator dashboard.

## 1 Network Packet Capture

Continuously captures packets from the selected network interface using Scapy in a dedicated background thread.

## 2 Packet Analysis

Extracts source IP, destination IP, ports, protocol type, TCP flags, and timestamp from each captured packet.

## 3 Port Scan Detection

Monitors connection attempts per source IP; multiple destination ports within a short time window trigger a port scan alert.

## 4 TCP SYN Flood Detection

Monitors SYN packet counts per source IP; a large number within a short interval triggers a SYN flood alert.

## 5 Threat Classification & Alert Generation

Classifies threats as **Low** (minor), **Medium** (repeated), or **High** (active attack). Alerts include source IP, destination IP, attack type, severity, and timestamp.

## 6 Event Logging & Dashboard

All detected events are stored for future analysis. Flask server processes IDS data and the real-time dashboard displays network security status, statistics, and historical logs.

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# Attack Detection & Conclusion

## Port Scan Detection

Monitors connection attempts from each source IP. Multiple destination ports contacted within a short period indicate possible port scanning. The system tracks connection attempts, identifies scanning patterns, and generates an alert when the threshold is exceeded.

## TCP SYN Flood Detection

Monitors TCP packets containing the SYN flag. A large number of SYN packets within a short interval indicates a possible SYN flood attack. The system counts SYN packets per source IP and generates an alert when the threshold is exceeded.

Both detection methods use **time-window analysis** to identify attack patterns in real time.

## Key Outcomes

### Real-Time Monitoring

Live packet capture and instant threat detection with no manual intervention required

### Severity-Aware Alerts

Low / Medium / High classification enables prioritized incident response

### Lightweight Deployment

Runs on standard hardware; no enterprise infrastructure needed

### Educational Value

Demonstrates TCP/IP stack, IP addressing, port analysis, and network security principles in a hands-on system

## Project Summary

This Network IDS successfully bridges the gap between enterprise-grade detection tools and the practical needs of small networks and academic environments — delivering real-time threat visibility through a clean, accessible web dashboard built entirely with open-source Python tooling.

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